

Named Standard IPv4 ACL Syntax

Access Control Lists | Network Security | Cisco Networking Academy

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Cisco Networking Academy | 2026

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Two Methods to Modify an ACL

- An ACL rarely stays untouched — new rules, new subnets, or a typo always turn up sooner or later
- Cisco IOS gives two ways to edit an ACL that's already configured on a router
- Method 1 rewrites the whole ACL outside the router, then pastes the fixed version back in
- Method 2 edits, inserts, or removes a single line using sequence numbers

Example:

A named ACL called NO_ACCESS can be fixed with either method — which one you pick just depends on how much of it actually needs to change.

Text Editor Method

Best for big rewrites. Copy the ACL out, edit it in a text editor, delete the old one, paste the new one back in.

Sequence Number Method

Best for quick fixes. Insert or delete one line by its sequence number — nothing else in the ACL moves.

Text Editor Method

- Run show running-config and copy the ACL lines out of the router
- Paste them into a plain text editor, like Notepad, and make the changes there
- Delete the existing ACL from the router first, or the new lines just get added to the end of the old one
- Paste the corrected version back into global configuration mode

Tip:

The moment the ACL is deleted, the router stops filtering that traffic altogether — so have the edited version ready to paste back in right away.

1 show running-config

2 edit the ACL in Notepad

3 no ip access-list standard NAME

4 paste the edited ACL back in

Sequence Number Method

- Cisco IOS numbers each line of a named ACL automatically, counting up by 10 — 10, 20, 30...
- `show access-lists` displays the ACL together with its current sequence numbers
- A new line slots in anywhere just by giving it a number that sits between two existing ones
- A single line is removed with “no” followed only by its sequence number — everything else stays put

Example:

Inserting sequence 15 between lines 10 and 20 adds a new rule without retyping the whole ACL.

```
10 deny host 192.168.1.1
```

```
15 deny host 192.168.1.2 ← new
```

```
20 permit any
```

The new line 15 fits neatly between 10 and 20, no renumbering needed.

Syntax Checker

- Syntax Checker lives inside the NetAcad course itself — no router or Packet Tracer required
- It uses the exact commands from this topic, so the practice matches the real CLI syntax
- Wrong syntax gets flagged immediately, before it's ever tried on real equipment
- A safe, low-pressure way to build muscle memory for both editing methods

Example:

Typing `ip access-list standard NO_ACCESS`, then `15 deny host 192.168.1.2`, inside Syntax Checker practices the exact steps used on a real router.



Practicing the syntax before it's graded

Summary

- Named ACLs can be edited two ways: a text editor or sequence numbers
- Text editor method = copy it out, edit it, delete the old one, paste the new one back in
- Sequence number method = insert or delete a single line without touching the rest
- show access-lists reveals the current line numbers, so you know exactly where a new rule will land
- Syntax Checker in NetAcad is a safe place to practice both methods before touching real hardware

References

Cisco Networking Academy

<https://www.netacad.com>

Cisco — Configuring IP Access Lists

<https://www.cisco.com/c/en/us/support/docs/ip/access-lists/26448-ACLsamples.html>

Cisco — IP Access List Entry Sequence Numbering

https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/sec_data_acl/configuration/xe-16-12/sec-data-acl-xe-16-12-book/sec-acl-seq-num.html

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<https://pixabay.com/photos/student-typing-keyboard-text-849826/>